

Optimization 1

ISE 5406

Lecture 8 Lab: From Fritz John to KKT to Duality

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Where We Are in the Course

Last Time (Lectures 6–7): KKT Conditions and Duality Theory

- KKT stationarity, complementary slackness, and optimality conditions
- The Lagrangian dual problem and dual function
- Weak duality theorem: $d^* \leq p^*$
- Strong duality and its role in convex optimization

Today (Lecture 8 Lab): From Fritz John to KKT to Duality

- Constraint qualifications (CQ): LICQ and MFCQ
- Fritz John conditions and when they guarantee KKT
- Hands-on exploration of CQ via worked examples
- Connection between KKT multipliers and dual optimizers (shadow prices)
- Hands-on: linking KKT solutions to dual problems

Next Time (Lecture 9): Pseudoconvex Functions and Generalizations

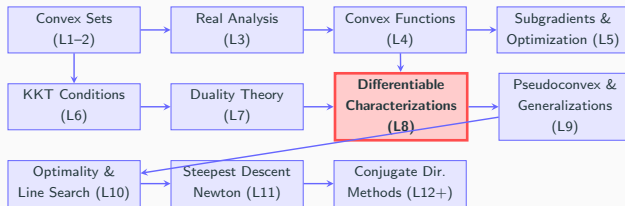
- Pseudoconvex and quasiconvex functions
- Generalized convexity classes and their properties
- Characterizations and relationships

Coming Soon (Lectures 10–11): Optimality and Algorithms

- Optimality conditions for non-smooth problems
- Line search methods and descent algorithms
- Gradient descent, Newton, and conjugate direction methods

Big Picture: This lab bridges the gap between the Fritz John conditions (which always exist) and the KKT conditions (which require a constraint qualification). We'll see that CQs are not pedantic—they ensure multiplier existence and uniqueness, and they unlock strong duality.

Big Picture: Course Roadmap



Why This Lab Exercise Matters

- Constraint qualifications are not abstract—they determine when KKT holds
- Understanding CQ is essential for recognizing solvable vs. problematic problem structures
- KKT multipliers have a dual interpretation: they are shadow prices, the rate of optimal value change w.r.t. constraint perturbations
- Hands-on experience reveals why LICQ and MFCQ guarantee multiplier existence and duality

Instructor Pacing Note: Structured Lab Exercise

Total Time: 62 minutes (fits 50-minute class with take-home)

Timing Breakdown:

- **Round 1** (Everything Works): 8 min work + 3 min discuss = 11 min
- **Round 2** (KKT Breaks): 8 min work + 5 min discuss = 13 min
- **Round 3** (Diagnosing CQ): 8 min work + 5 min discuss = 13 min
- **Round 4** (Uniqueness and MFCQ): 5 min work + 3 min discuss = 8 min
- **Round 5** (KKT to Duality): 10 min work + 7 min discuss + wrap-up = 17 min

For 50-minute class: Do Rounds 1–4 in class (45 min), assign Round 5 as take-home. Complete pacing allows deeper discussion.

Group Size: Assign 2–3 people per round. Rotate roles: one solves, one explains, one records.

Quick Reference: Fritz John vs. KKT

Fritz John Conditions:

Always exist at any local optimum, even when KKT fails.

$$\lambda_0 \nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) = 0$$

$$\lambda_i g_i(x^*) = 0 \text{ (complementary slackness)}$$

$$\lambda_i \geq 0 \text{ for all } i$$

$$\lambda_0 \geq 0$$

$$(\lambda_0, \lambda_1, \dots, \lambda_m) \neq (0, 0, \dots, 0)$$

The key point: λ_0 may be zero, making the objective “invisible”.

KKT Conditions:

A special case when $\lambda_0 > 0$.

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) = 0$$

$$\lambda_i g_i(x^*) = 0 \text{ (complementary slackness)}$$

$$\lambda_i \geq 0 \text{ for all } i$$

Can always normalize $\lambda_0 = 1$ in KKT.

Key question: When is $\lambda_0 > 0$ guaranteed? $\Rightarrow$ Constraint Qualifications (CQ)

Quick Reference: Constraint Qualifications (CQ)

Linear Independence CQ (LICQ)

The gradients of active constraints at x^* are linearly independent:

$$\{\nabla g_i(x^*) : i \in \mathcal{A}(x^*)\} \text{ are linearly independent}$$

where $\mathcal{A}(x^*) = \{i : g_i(x^*) = 0\}$ is the active set.

Strongest CQ; guarantees unique multipliers.

Mangasarian–Fromovitz CQ (MFCQ)

Active constraint gradients span a proper subspace: there exists a direction d such that

$$\nabla g_i(x^*)^T d < 0 \quad \text{for all } i \in \mathcal{A}(x^*)$$

(strictly negative on all active constraints).

Weaker than LICQ; allows multiplier non-uniqueness but forces $\lambda_0 > 0$.

Hierarchy: LICQ $\Rightarrow$ MFCQ $\Rightarrow$ ($\lambda_0 > 0$ in Fritz John) $\Rightarrow$ KKT holds.

Outline

Round 1: Everything Works

Round 2: KKT Breaks — Fritz John Works

Round 3: Diagnosing the Problem — Constraint Qualifications

Round 4: Uniqueness and Non-Uniqueness

Round 5: KKT to Duality

The Complete Picture

Round 1 Exercise: Everything Works

A Simple Constrained Problem

Problem:

$$\min (x_1 - 2)^2 + (x_2 - 2)^2 \quad \text{s.t.} \quad g_1 : x_1 + x_2 \leq 2, \quad g_2 : x_1 \geq 0, \quad g_3 : x_2 \geq 0$$

Tasks (8 minutes):

1. Find the unconstrained minimizer and check if it is feasible.
2. Write the KKT stationarity condition:
$$\nabla f(x^*) + \lambda_1 \nabla g_1(x^*) + \lambda_2 \nabla g_2(x^*) + \lambda_3 \nabla g_3(x^*) = 0.$$
3. Solve the KKT system to find x^* , λ_1^* , λ_2^* , λ_3^* , and $f(x^*)$.
4. Check the LICQ condition: are the active constraint gradients linearly independent?

Round 1 Solution

Analysis

Step 1: Unconstrained minimizer:

$$\nabla f(x_1, x_2) = (2(x_1 - 2), 2(x_2 - 2))^T = 0 \text{ gives } (x_1, x_2) = (2, 2).$$

Check feasibility: $g_1(2, 2) = 4 > 0$ **INFEASIBLE**. The constraint g_1 is binding at optimum.

Step 2 & 3: At $x^* = (1, 1)$, only g_1 is active. KKT stationarity gives:

$$\begin{aligned} \begin{pmatrix} 2(1-2) \\ 2(1-2) \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} -2 \\ -2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \Rightarrow \lambda_1 &= 2 \end{aligned}$$

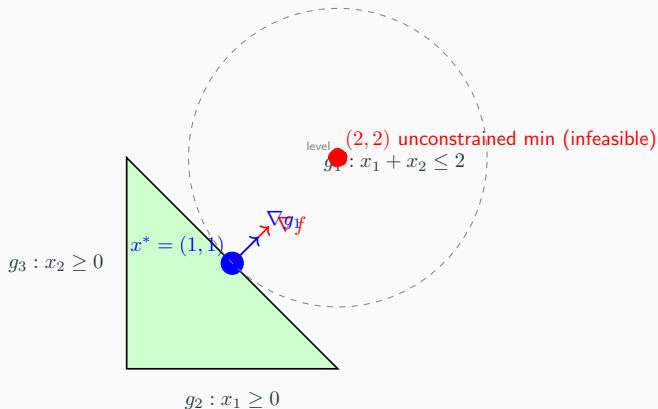
Complementary slackness: $\lambda_2, \lambda_3 = 0$ (since $g_2, g_3 < 0$). Thus $\lambda^* = (2, 0, 0)$.

Objective value: $f(1, 1) = (1 - 2)^2 + (1 - 2)^2 = 2$.

Step 4: Active constraint gradient at $x^* = (1, 1)$:

$\nabla g_1(1, 1) = (1, 1)^T \neq 0$. **LICQ holds** (one non-zero vector is linearly independent). Therefore **multipliers are unique**.

Round 1: Geometric Insight



At $x^* = (1, 1)$, the gradient ∇f is parallel to the active constraint gradient ∇g_1 . The multiplier $\lambda_1 = 2$ is the scaling factor.

Round 2 Exercise: KKT Breaks

A Problem Where KKT Fails

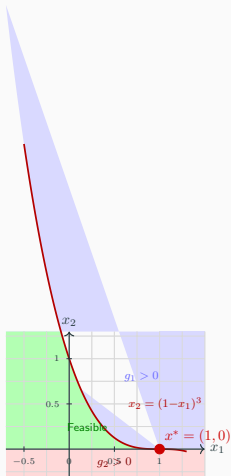
Problem:

$$\min -x_1 \quad \text{s.t.} \quad g_1 : x_2 - (1 - x_1)^3 \leq 0, \quad g_2 : -x_2 \leq 0$$

Tasks (8 minutes):

1. Find the optimal point x^* . (Hint: the objective wants x_1 as large as possible.)
2. Compute the gradients $\nabla f(x^*)$, $\nabla g_1(x^*)$, $\nabla g_2(x^*)$ at x^* .
3. Try to solve the KKT stationarity equation. What goes wrong?
4. Now try Fritz John: $\lambda_0 \nabla f(x^*) + \lambda_1 \nabla g_1(x^*) + \lambda_2 \nabla g_2(x^*) = 0$ with $\lambda_0, \lambda_1, \lambda_2 \geq 0$, not all zero. Does it have a solution?

Round 2: Problem Geometry



The feasible region (green) is bounded by the red curve and the x-axis.

The optimum $x^* = (1,0)$ maximizes x_1 .

Round 2 Solution

Why KKT Fails; Fritz John Succeeds

Step 1: The objective $f(x_1, x_2) = -x_1$ wants to maximize x_1 . Within the feasible region (below the curve, above the x-axis), x_1 reaches its maximum at $(1, 0)$, where the curve meets the x-axis.

Step 2: Compute gradients at $x^* = (1, 0)$:

$$\nabla f(1, 0) = (-1, 0)^T$$

$$\nabla g_1(1, 0) = \frac{\partial}{\partial x_1}(x_2 - (1 - x_1)^3)\big|_{(1,0)} = 3(1 - x_1)^2\big|_{(1,0)} = 0, \quad \frac{\partial}{\partial x_2} = 1$$

$$\Rightarrow \nabla g_1(1, 0) = (0, 1)^T$$

$$\nabla g_2(1, 0) = (0, -1)^T$$

Step 3: KKT stationarity fails: We need

$$(-1, 0)^T + \lambda_1(0, 1)^T + \lambda_2(0, -1)^T = 0. \text{ First component: } -1 = 0.$$

CONTRADICTION.

Step 4: Fritz John succeeds: Allow $\lambda_0 \geq 0$:

$$\lambda_0(-1, 0)^T + \lambda_1(0, 1)^T + \lambda_2(0, -1)^T = (0, 0)^T$$

Setting $\lambda_0 = 0$, $\lambda_1 = \lambda_2 = 1$ gives $(0, 1)^T + (0, -1)^T = 0$. **Check!** Not all zero: **Fritz John holds.**

Round 2: What Happened?

Key Insight

When $\lambda_0 = 0$ in Fritz John, the objective gradient $\nabla f(x^*)$ is *absent* from the optimality condition. The constraint gradients $\nabla g_1(1, 0) = (0, 1)^T$ and $\nabla g_2(1, 0) = (0, -1)^T$ are *opposite directions*—they are linearly dependent.

Fritz John just says: “The active constraint gradients are linearly dependent.” It does **not** involve the objective. KKT requires that the objective participates ($\lambda_0 > 0$), which fails here because the constraints alone can balance.

The Question

How do we guarantee $\lambda_0 > 0$, so that KKT always holds? What is the condition that says: “Even if constraint gradients are dependent, they cannot perfectly balance without the objective”?

Answer: This is where *constraint qualifications* come in.

Round 3 Exercise: Diagnosing with Constraint Qualifications

Understanding Why CQ Fails

Round 1 vs Round 2 Comparison:

1. Fill in the table below (comparing active constraint gradients in Rounds 1 and 2).
2. **Key argument:** Suppose MFCQ holds: there exists d such that $\nabla g_i(x^*)^T d < 0$ for all active i . Now suppose (for contradiction) that Fritz John has $\lambda_0 = 0$. Then:

$$\sum_{i \in \mathcal{A}} \lambda_i \nabla g_i(x^*) = 0$$

Take inner product with d :

$$\sum_{i \in \mathcal{A}} \lambda_i [\nabla g_i(x^*)^T d] = 0$$

Why does this lead to a contradiction? (Hint: each term is (non-negative) $\times$ (negative)...))

3. Verify that MFCQ fails in Round 2: find why no direction d can satisfy $\nabla g_i(x^*)^T d < 0$ for both $i = 1, 2$.

Round 3: Comparison Table

Constraint Qualification Analysis

Property	Round 1	Round 2
Active constraint at x^*	g_1 only	g_1, g_2 both
Active gradients	$\nabla g_1 = (1, 1)^T$	$\nabla g_1 = (0, 1)^T, \nabla g_2 = (0, -1)^T$
Linearly independent?	YES (LICQ holds)	NO (opposite directions)
MFCQ direction d	$d = (-1, -1)$ works $\nabla g_1^T d = -2 < 0$	None exist Need $d_2 < 0$ AND $d_2 > 0$
KKT exists?	YES	NO
λ_0 from Fritz John	> 0 (can set $\lambda_0 = 1$)	$= 0$ (forced)
Multiplier λ_1	Unique: $\lambda_1 = 2$	Non-existent in KKT

Round 3: The Key Proof

Claim: $\text{MFCQ} \Rightarrow \lambda_0 > 0$ in Fritz John

Proof by Contradiction:

Suppose MFCQ holds at x^* , and suppose (for contradiction) that Fritz John has $\lambda_0 = 0$. Then:

$$\sum_{i \in \mathcal{A}(x^*)} \lambda_i \nabla g_i(x^*) = 0 \quad \text{with } \lambda_i \geq 0, \text{ not all zero}$$

By MFCQ, there exists direction d with $\nabla g_i(x^*)^T d < 0$ for all $i \in \mathcal{A}(x^*)$.

Take inner product of the Fritz John equation with d :

$$\sum_{i \in \mathcal{A}} \lambda_i [\nabla g_i(x^*)^T d] = 0$$

Each term analysis:

- $\lambda_i \geq 0$ (non-negative, ≥ 0)
- $\nabla g_i(x^*)^T d < 0$ (negative)
- Product: $\lambda_i [\nabla g_i(x^*)^T d] \leq 0$

Since at least one $\lambda_i > 0$ (not all zero) and each product ≤ 0 , with at least one **strictly** < 0 :

$$\sum_{i \in \mathcal{A}} \lambda_i [\nabla g_i(x^*)^T d] < 0$$

This contradicts the sum equaling 0. Therefore $\lambda_0 > 0$. $\square$

Round 4 Exercise: Uniqueness of Multipliers

When Are Multipliers Unique?

Problem:

$$\min x_1 \quad \text{s.t.} \quad g_1 : -x_1 \leq 0, \quad g_2 : -2x_1 \leq 0$$

Tasks (5 minutes):

1. Find the optimal point x^* .
2. Check LICQ: are the active constraint gradients linearly independent?
3. Check MFCQ: does there exist a direction d with $\nabla g_i(x^*)^T d < 0$ for all active i ?
4. Find **all** KKT multiplier pairs (λ_1, λ_2) satisfying stationarity. Are they unique?
5. **Bonus:** Explain why LICQ guarantees uniqueness while MFCQ does not.

Round 4 Solution

Understanding Non-Unique Multipliers

Step 1: The objective is $f(x_1) = x_1$. Minimizing over feasible set (where $x_1 \leq 0$ and $x_1 \leq 0$, i.e., $x_1 \leq 0$) gives $x^* = 0$.

Step 2: Both constraints are active at $x^* = (0)$:

$$\nabla g_1(0) = (-1, 0)^T, \quad \nabla g_2(0) = (-2, 0)^T$$

These are parallel (proportional). **LICQ fails.**

Step 3: Check MFCQ. We need $d = (d_1, d_2)$ with:

$$\nabla g_1(0)^T d = -d_1 < 0 \Rightarrow d_1 > 0$$

$$\nabla g_2(0)^T d = -2d_1 < 0 \Rightarrow d_1 > 0$$

Both require $d_1 > 0$. Choose $d = (1, 0)$. **MFCQ holds.**

Step 4: KKT stationarity: $\nabla f(0) + \lambda_1 \nabla g_1(0) + \lambda_2 \nabla g_2(0) = 0$:

$$(1, 0)^T + \lambda_1(-1, 0)^T + \lambda_2(-2, 0)^T = 0$$

$$\Rightarrow 1 - \lambda_1 - 2\lambda_2 = 0$$

$$\Rightarrow \lambda_1 + 2\lambda_2 = 1$$

Infinitely many solutions: $(\lambda_1, \lambda_2) = (1 - 2t, t)$ for $t \in [0, 1/2]$.

Multipliers are non-unique.

Step 5: LICQ makes constraint gradients a basis $\Rightarrow$ the coefficients (multipliers) are uniquely determined. Without LICQ, gradients are dependent $\Rightarrow$ underdetermined system $\Rightarrow$ multiple solutions.

Part I Summary: CQ Hierarchy

Comparison of All Three Rounds

Property	Round 1	Round 4	Round 2
LICQ?	YES	NO	NO
MFCQ?	YES	YES	NO
KKT exists?	YES	YES	NO
Multipliers unique?	YES	NO	N/A

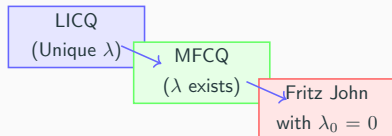
The CQ Hierarchy

$\text{LICQ} \Rightarrow \text{MFCQ} \Rightarrow (\lambda_0 > 0) \Rightarrow \text{KKT Exists}$

Unique multipliers

vs

Multipliers exist



Round 5 Exercise: KKT to Duality Connection

Building and Solving the Dual Problem

Problem (Same as Round 1):

$$\min (x_1 - 2)^2 + (x_2 - 2)^2 \quad \text{s.t.} \quad g_1 : x_1 + x_2 \leq 2$$

(suppress non-negativity for simplicity; they become inactive)

Known results from Round 1: $x^* = (1, 1)$, $\lambda^* = 2$, $p^* = 2$.

Tasks (10 minutes):

1. Write the Lagrangian:

$$L(x_1, x_2, \lambda) = (x_1 - 2)^2 + (x_2 - 2)^2 + \lambda(x_1 + x_2 - 2).$$

2. Minimize L over x (i.e., solve $\frac{\partial L}{\partial x_i} = 0$). Express $x_1(\lambda)$ and $x_2(\lambda)$.
3. Derive the dual function $g(\lambda) = \min_{x_1, x_2} L(x_1, x_2, \lambda)$.
4. Maximize the dual function: find λ_{dual}^* and $d^* = g(\lambda_{\text{dual}}^*)$.
5. Compare: Is $p^* = d^*$? What does this mean? (Hint: This is “strong duality”)

Round 5 Solution: Building the Dual

Constructing the Dual Function

Step 1 & 2: The Lagrangian is:

$$L(x_1, x_2, \lambda) = (x_1 - 2)^2 + (x_2 - 2)^2 + \lambda(x_1 + x_2 - 2)$$

Minimize over x :

$$\frac{\partial L}{\partial x_1} = 2(x_1 - 2) + \lambda = 0 \Rightarrow x_1(\lambda) = 2 - \frac{\lambda}{2}$$

$$\frac{\partial L}{\partial x_2} = 2(x_2 - 2) + \lambda = 0 \Rightarrow x_2(\lambda) = 2 - \frac{\lambda}{2}$$

Step 3: Substitute back into L :

$$\begin{aligned} g(\lambda) &= \left(2 - \frac{\lambda}{2} - 2\right)^2 + \left(2 - \frac{\lambda}{2} - 2\right)^2 + \lambda \left(2 \cdot \left(2 - \frac{\lambda}{2}\right) - 2\right) \\ &= 2 \cdot \frac{\lambda^2}{4} + \lambda(4 - \lambda - 2) \\ &= \frac{\lambda^2}{2} + \lambda(2 - \lambda) \\ &= \frac{\lambda^2}{2} + 2\lambda - \lambda^2 \\ &= -\frac{\lambda^2}{2} + 2\lambda \end{aligned}$$

Dual function: $g(\lambda) = 2\lambda - \frac{\lambda^2}{2}$ (a concave quadratic in λ).

Round 5 Solution: Solving the Dual

Maximizing the Dual Function

Step 4: Maximize $g(\lambda) = 2\lambda - \frac{\lambda^2}{2}$ over $\lambda \geq 0$:

$$\frac{dg}{d\lambda} = 2 - \lambda = 0 \Rightarrow \lambda_{\text{dual}}^* = 2$$

At $\lambda = 2$:

$$d^* = g(2) = 2 \cdot 2 - \frac{2^2}{2} = 4 - 2 = 2$$

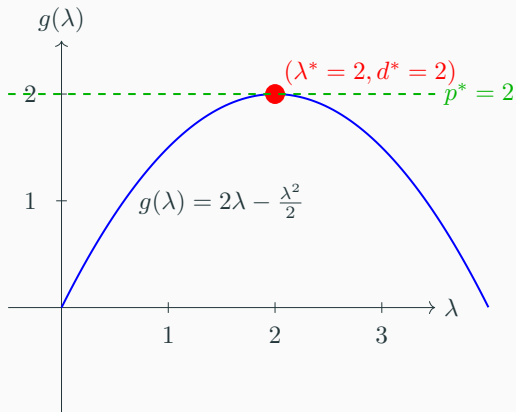
Step 5: Compare primal and dual:

$$p^* = 2 \quad \text{and} \quad d^* = 2$$

Strong Duality holds: $p^* = d^*$, i.e., zero duality gap!

Moreover, the optimal λ from the dual ($\lambda_{\text{dual}}^* = 2$) **matches** the KKT multiplier from Round 1 ($\lambda^* = 2$). **They are the same object!**

Round 5: Dual Function Visualization



The dual function $g(\lambda)$ is concave. Its maximum at $\lambda^* = 2$ achieves $d^* = p^* = 2$ (strong duality).

Round 5: Why Does Strong Duality Hold?

Weak Duality (Always True)

For any feasible x and any $\lambda \geq 0$:

$$g(\lambda) = \min_x L(x, \lambda) \leq L(x, \lambda) \leq f(x) + \lambda \cdot 0 = f(x) \leq p^*$$

(by complementary slackness at optimum). Thus $d^* = \max_{\lambda} g(\lambda) \leq p^*$.

Strong Duality (Requires a Constraint Qualification)

Strong duality $d^* = p^*$ holds when a **constraint qualification** is satisfied.

Slater's Condition: For convex problems, if $\exists$ strictly feasible $\hat{x}$ (i.e., $g_i(\hat{x}) < 0$ for all i), then strong duality holds.

In Round 1: Check $\hat{x} = (0, 0)$. We have $g_1(0, 0) = 0 + 0 - 2 = -2 < 0$.

Slater holds!

Key connection: Slater $\Rightarrow$ MFCQ $\Rightarrow$ KKT exists $\Leftrightarrow$ strong duality. The multipliers are not just mathematical artifacts—they are the dual optimizers!

Round 5: The KKT-Duality Connection

At a KKT Point (x^*, λ^*) :

1. **Stationarity** $\Rightarrow x^*$ minimizes $L(\cdot, \lambda^*) \Rightarrow g(\lambda^*) = L(x^*, \lambda^*)$.
2. **Complementary Slackness** $\Rightarrow \lambda_i^* g_i(x^*) = 0 \Rightarrow$
 $L(x^*, \lambda^*) = f(x^*) + \sum_i \lambda_i^* \cdot 0 = f(x^*)$.
3. **Combining (1) and (2):** $g(\lambda^*) = L(x^*, \lambda^*) = f(x^*) = p^*$.

Therefore, $g(\lambda^*) = p^*$, i.e., the dual is optimized at λ^* .

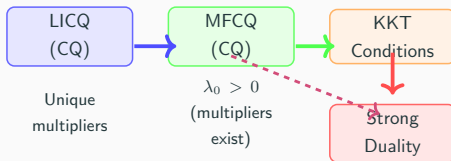
The Complete Picture:

KKT multipliers = dual optimizers = shadow prices

They are the same object viewed from three different angles:

1. **Primal view:** Lagrange multipliers solving the KKT system.
2. **Dual view:** Optimizers of the dual problem.
3. **Economic view:** Shadow prices—the rate of change of optimal value w.r.t. constraint perturbations.

The Complete Picture: CQ, KKT, and Duality



Key Results

- LICQ (Active gradients linearly independent) $\Rightarrow$ MFCQ $\Rightarrow$ KKT conditions hold
- MFCQ $\Rightarrow \lambda_0 > 0$ in Fritz John, so KKT is just Fritz John with $\lambda_0 = 1$
- For convex problems: KKT $\Leftrightarrow$ optimal (first-order sufficient condition)
- For convex problems: Slater's condition $\Rightarrow$ MFCQ and strong duality
- At KKT point: multipliers are dual optimizers and shadow prices

Key Takeaways: The Main Results

Fritz John: The Universal Condition

At any local optimum x^* of a smooth problem, there exist

$(\lambda_0, \lambda_1, \dots, \lambda_m)$ with $\lambda_0, \lambda_i \geq 0$, not all zero, satisfying:

$$\lambda_0 \nabla f(x^*) + \sum_i \lambda_i \nabla g_i(x^*) = 0, \quad \lambda_i g_i(x^*) = 0$$

But λ_0 might be zero, making the objective invisible!

Constraint Qualifications Guarantee KKT

- **LICQ** (Strongest): Active constraint gradients linearly independent
 $\Rightarrow$ unique multipliers
- **MFCQ** (Weaker): Exists direction strictly decreasing all active constraints $\Rightarrow \lambda_0 > 0$ in Fritz John
- Both $\Rightarrow$ KKT conditions are necessary for optimality

Duality Connection

For convex problems with Slater condition:

- **KKT multipliers = dual optimizers = shadow prices**
- Strong duality holds: $p^* = d^*$ (zero duality gap)
- Multipliers measure rate of change of optimal value w.r.t. constraint slack

Preview: Pseudoconvex Functions and Generalizations

- Quasiconvex and pseudoconvex functions
- When do generalized convexity classes preserve optimality characterizations?
- Relationships between LICQ, MFCQ, and generalizations
- Building toward a toolkit for analyzing non-convex problems

Questions?

Office hours: (Days/Times TBD). Post questions on course forum or email.